DS5020
Mini Project 1 Memo
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We decided the best method for this task is truncated Singular Value Decomposition (SVD). We compared it with a popular alternative, Principle Component Analysis (PCA), and determined that SVD would be both more efficient and effective. The key difference between the two methods is that PCA centers the data before decomposing. Considering how most customers only buy a few products, our data consists of mostly zeros. In theory, SVD should require less storage, since it preserves the sparse nature of the data. In reality, modern optimizations allowed both methods to reduce the size of the data from 156 MB to about 6.7 MB. However, SVD is still significantly faster to compute, taking only 0.27 seconds where PCA takes 0.43 seconds. For this dataset, the difference is nonconsequential, but when scaled up it could save hours of compute time.

Both of these methods allow us to select a parameter that determines how much of the information is kept. To decide on what value to use, we optimized for efficacy over a test group that was excluded from the data. We defined a ‘hit’ as the engine correctly including a customer’s future purchase based solely on prior purchases. For this metric, we only looked at new purchases, as it would be trivial to predict a recurring purchase. We found that the best value (k=100) resulted in an average hit rate of 28.62%, while maintaining enough information to reconstruct 69.6% of what was present initially. This is slightly better than both PCA and a basic best seller list, which have hit rates of 27.52% and 27.00%.

To split the customer base into groups as requested, we ran k-means clustering on the same data generated by SVD. K-means allows us to select any number of groups we want, so we compared how cleanly the data fit into different amounts of groups. It seemed like there was no clear cut number of groups, so we decided on six to give variability without being overwhelming. Within each of these groups, we provided information about the amount of unique purchases, where the customers are located, and the most popular products.

There are some possible sources of error in the recommendation engine and in our assessment of it that should be noted. First, we evaluated based on the most recent two months of data, which happened to include winter holidays. This likely changed customer’s purchase decisions, which may have affected the relative hit rate for each method. Also, the hit rates were all very close, so it is possible that a different test split could result in a different method being recommended. An inherent flaw of recommendation based on prior purchases is that we will not have enough information on new customers to make specific predictions. They will likely be recommended best sellers until they have established preferences. Personal information has been appropriately handled by replacing all customer IDs with new randomized IDs that are disconnected from any personal information.